



PRODUCT NAME

High-performance Hair Oil Formula

Description

A liquid formulation designed to provide a moisturizing, non-greasy finish with nourishing properties ideal for all hair types.

Technical Specifications

pH Level:	7
Viscosity:	Low
Shelf Life:	24 months
Batch Size:	10-500 liters
Processing Time:	2 hours
Temperature:	Room temperature for mixing; 40°C for oil blending
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing vessel, heating source, stirrer, funnel
Certification:	Made with GMP compliance

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Sunflower Seed Oil	Helianthus Annuus (Sunflower) Seed Oil	30.0%	Base oil, moisturizing
2	Argan Oil	Argania Spinosa Kernel Oil	20.0%	Nourishing oil, adds shine
3	Coconut Oil	Cocos Nucifera (Coconut) Oil	15.0%	Moisturizing, provides a light, non-greasy feel
4	Almond Oil	Prunus Amygdalus Dulcis (Sweet Almond) Oil	10.0%	Conditioning, nourishing
5	Squalane	Squalane	10.0%	Non-greasy emollient, enhances absorption
6	Vitamin E	Tocopheryl Acetate	2.0%	Antioxidant, extends shelf life
7	Fragrance	Fragrance	1.0%	Adds pleasant scent
8	Polysorbate 80	Polysorbate 80	8.0%	Solubilizes fragrance, non-irritant
9	Phenoxyethanol	Phenoxyethanol	1.0%	Preservative
10	Water	Aqua	3.0%	Diluent, maintains low viscosity
Total			100%	

Manufacturing Process

Phase 1: Phase A

- Combine Sunflower Seed Oil, Argan Oil, Coconut Oil, Almond Oil, and Squalane in a mixing vessel.
- Heat gently to 40°C and mix until homogenous.

Phase 2: Phase B

1. In a separate vessel, mix Vitamin E, Phenoxyethanol, Polysorbate 80, and Fragrance.
2. Add the Phase B mixture to Phase A slowly, stirring continuously.

Phase 3: Phase C

1. Add water to the combined mixture and stir until fully integrated.
2. Cool the mixture to room temperature while continuously stirring.
3. Transfer the product into final containers.

Required Equipment

Mixing vessel, heating source, stirrer, funnel

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Made with GMP compliance

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 10-500 liters.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a few drops of the hair oil to the palm of your hand. Spread evenly throughout damp or dry hair, avoiding the roots. Style as desired.